

Adithya Satheesh

Robotics Engineer – Autonomous Systems & Embedded AI
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Professional Summary

Robotics Engineer with over a year of hands-on experience in autonomous systems, multi-sensor fusion, embedded systems, and real-time computer vision pipelines. Proven track record in developing scalable autonomy stacks for GPS-denied environments and integrating complex hardware-software systems. Expert in ROS2 architecture, state estimation, and deploying lightweight machine learning models at the edge.

Technical Skills

Programming Languages: C++, Python, MATLAB, SQL

Robotics Ecosystem: ROS2, TF2, Nav2, Sensor Fusion, State Estimation, SLAM Concepts

Computer Vision & ML: YOLO (v5/v8), OpenCV, RGB-D Processing, Point Cloud Processing, Edge AI, TinyML

Embedded Systems: Teensy 4.1, Arduino Nano 33 BLE, ESP32, Motor Control (PWM), Encoders, Serial Protocols

Tools & Frameworks: Linux (Ubuntu), Git, CMake, PlatformIO, Docker

Mathematics & Control: Linear Algebra, State-Space Modeling, Kalman Filtering (EKF), Kinematics

Professional Experience

National Aerospace Laboratories (NAL)

Bengaluru, India

Project Associate – Unmanned Ground Vehicles & Embedded Systems

2024 – Present

- Design and implement advanced perception and navigation stacks for experimental Unmanned Ground Vehicles (UGVs) specialized in automated aircraft structural inspection.
- Develop real-time multi-sensor data acquisition interfaces, synchronizing critical telemetry data across embedded microcontrollers and high-level ROS2 computing nodes.
- Architect custom computer vision pipelines utilizing YOLO and OpenCV to optimize automated target tracking and safety-critical defect localization.
- Lead full-stack hardware-software integration, system-level debugging, and bench testing, reducing overall localization drift on hardware prototypes.

Selected Projects

Multi-Sensor Fusion for Autonomous UGV Localization

Technologies: ROS2, C++, EKF, VINS-Fusion, LiDAR, Intel RealSense, TF2

- Integrated visual-inertial odometry (VINS-Fusion) with 2D LiDAR and RGB-D depth sensing to build a robust localization stack for completely GPS-denied configurations.
- Designed and deployed an Error-State Extended Kalman Filter (ES-EKF) for sub-centimeter-level multi-sensor state estimation.
- Developed custom measurement models to execute LiDAR-based pose corrections and high-density depth projections onto local occupancy maps.
- Resolved complex spatio-temporal coordinate transformations and sensor synchronization bottlenecks using ROS2 TF2.
- Evaluated trajectory performance metrics quantitatively using Absolute Trajectory Error (ATE) and Relative Pose Error (RPE), paving the system-level migration pathway toward Factor-Graph Optimization.

Automated Resin Flow Infusion Tracking System

Technologies: Python, OpenCV, Digital Image Processing, Sensor Data Modeling

- Engineered a specialized image processing software suite to continuously track the front-line advancement of resin liquids along a metric manometer during aerospace composite manufacturing.
- Implemented custom color-space thresholding, dynamic region-of-interest (ROI) masking, and morphological filtering to ensure highly accurate boundary line segmentation under varying illumination profiles.
- Formulated mathematical volume-to-mass translation models to compute the instantaneous weight and volumetric distribution of total resin infused in real time, drastically improving precision control during material curation.

YOLO-Based Real-Time Defect Detection with Sensor Integration

Technologies: Python, ROS2, OpenCV, YOLOv8, Embedded Systems, Microcontrollers

- Built an end-to-end real-time object and surface defect detection pipeline by embedding a fine-tuned YOLOv8 network directly within a custom ROS2 perception architecture.
- Fused spatial ultrasonic distance sensor arrays with camera pixel bounding-box boundaries to dynamically compute the true 3D dimensions of targeted physical structures.
- Designed an optimized, fault-tolerant serial communication protocol establishing robust high-speed telemetry exchanges between the edge microcontroller and high-level ROS2 ecosystem nodes.
- Implemented advanced moving-average and Kalman-based measurement stabilization algorithms to suppress sensor noise and lower false-positive telemetry spikes.

Autonomous Mobile Robot with Multi-Motor Control

Technologies: Teensy 4.1, ROS2, Teleop, PWM Control, Encoders, Differential Drive

- Designed and prototyped a 4-wheel-drive mobile robot base leveraging high-torque continuous rotation servos regulated by a Teensy 4.1 microcontroller.
- Programmed microsecond-level precision PWM timing registries featuring finely calibrated neutral deadband windows to maximize operational motor response linearity.
- Developed a native ROS2 teleoperation interface to handle low-latency kinematic velocity commands while continuously broadcasting multi-channel encoder odometry ticks.

TinyML-Based Impact Localization System

Technologies: Arduino Nano 33 BLE Sense, Signal Processing, Edge ML, TensorFlow Lite

- Developed a real-time structural health monitoring solution to localize physical impacts on structured geometric grids using distributed Piezoelectric (PZT) sensor arrays.
- Authored on-board Digital Signal Processing (DSP) algorithms to extract key statistical time-series features directly inside constrained bare-metal hardware registers.
- Trained and compressed a lightweight machine learning classifier via TensorFlow Lite for Microcontrollers, maximizing inference efficiency for standalone hardware deployments.

Education

B.Tech in Robotics and Automation Engineering

First Class with Distinction

Graduated: 2024

Research Interests

Autonomous Navigation, Multi-Sensor Fusion, Visual-Inertial Odometry, SLAM, Embedded AI, Deep Learning for Computer Vision, Intelligent Robotic System Architectures.